

Radio Frequency Engineering

Laboratory report

SS2026

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1 COAXIAL RF LINE

1.1 Task description

This task introduces noise analysis of transistor and operational-amplifier circuits using Cadence PSpice. Following were the main sub-objectives:

- Noise analysis of a single stage emitter circuit
- An OpAmp noise model check (where Gain = +1)
- Noise analysis of inverting and non-inverting amplifier circuits (with Gain = 10)

1.2 Schematics

Figure 1.1 shows the circuit configuration of a single stage emitter circuit.

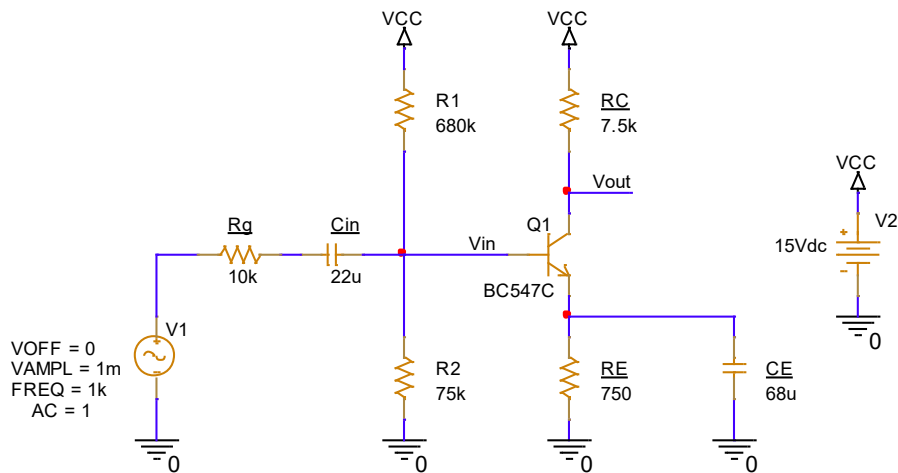


Figure 1.1: PSpice Schematic of a single stage emitter circuit with BC547C.

Figure 1.2 shows the circuit configuration of a non-inverting amplifier circuit.

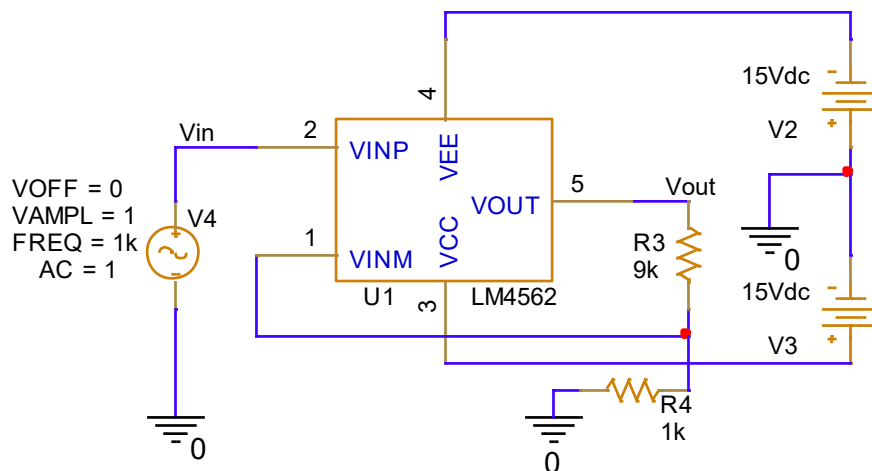


Figure 1.2: Schematic of a non-inverting amplifier with LM4562 (Gain = +10).

Figure 1.3 shows the circuit configuration of an inverting amplifier circuit.

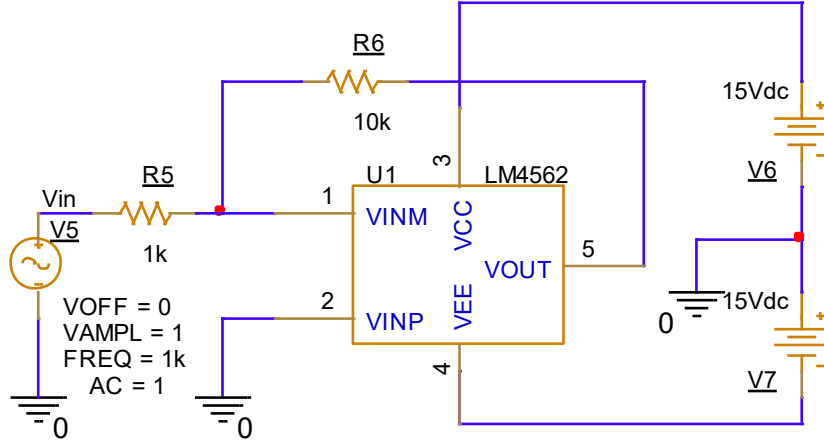


Figure 1.3: Schematic of an inverting amplifier with LM4562 (Gain = -10).

1.3 Table with measurement results

Table 1.1 shows the value of the simulated Input Noise Density and the value in the data sheet for the same.

Adjusted	Measured	Datasheet
f / Hz	$e_n / \text{nV}/\sqrt{\text{Hz}}$	$e_n / \text{nV}/\sqrt{\text{Hz}}$
10	6.810	6.4 [1]
1.0k	2.752	2.7 [1]

Table 1.1: Values corresponding to the OpAmp noise model check at 10 Hz and 1.0 kHz (Gain = +1).

1.4 Simulation curves and diagrams

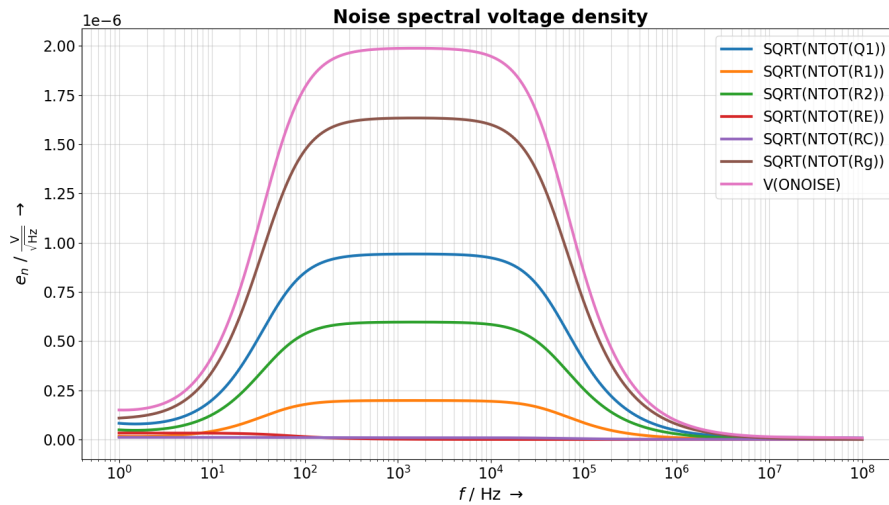


Figure 1.4: Simulation of the noise spectral voltage density of the emitter circuit ($\hat{u}=1\text{mV}$).

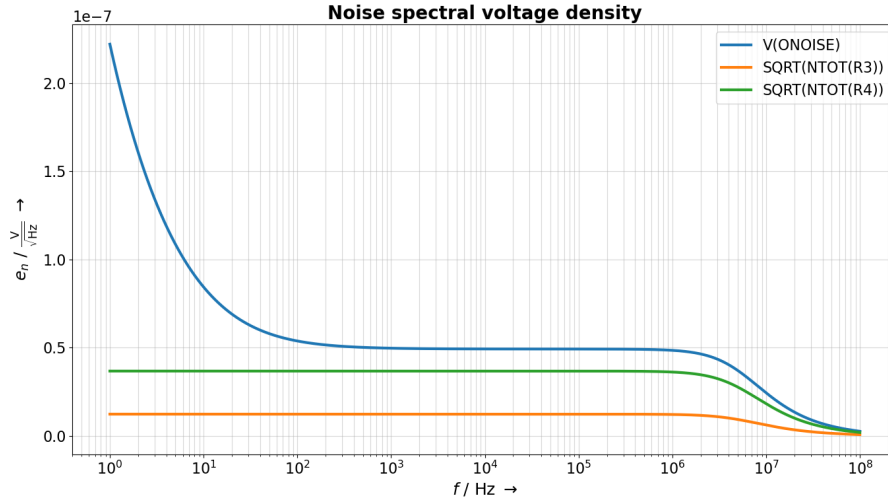


Figure 1.5: Simulation result showing the Noise spectral voltage density of a non-inverting amplifier circuit.

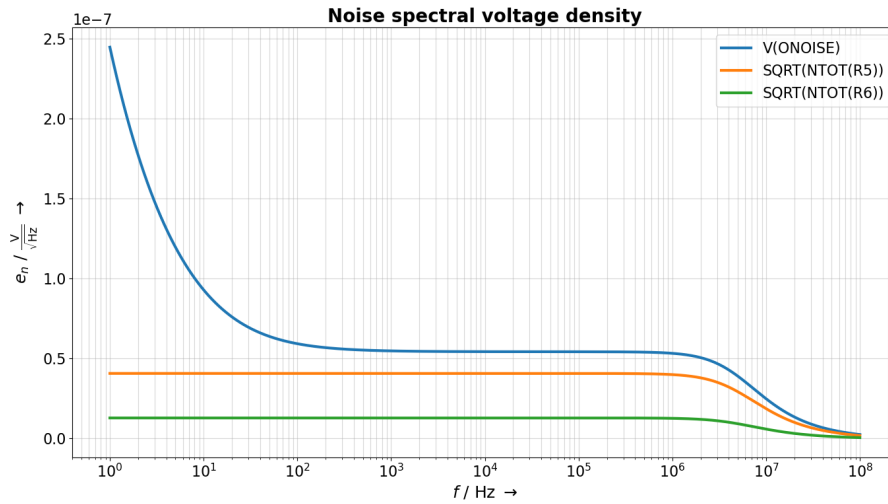


Figure 1.6: Simulation result showing the Noise spectral voltage density of an inverting amplifier circuit.

1.5 Discussion

- **Noise analysis of a single stage emitter circuit:**

Noise analysis of the single stage emitter circuit (figure 1.4) reveals that R_g contributes most to the total noise (brown trace).

- **OpAmp noise model check:**

A simple voltage follower circuit was built with LM4562 and using cursors, the simulated noise density e_n was read at 10 Hz as well as 1.0 kHz and compared to the datasheet values (see table 1.1).

- **Noise analysis of inverting and non-inverting amplifier circuits:**

The simulated output noise spectral density for these circuits with gain-10 amplifiers is shown in (figures 1.5 and 1.6). The spectrum shows the expected behaviour: increased low-frequency noise due to flicker noise and an approximately flat mid-band (white) noise floor. The change in slope marks the $1/f$ corner frequency.

2 Time Domain Reflectometry

2.1 Measurement A: Rise- and fall time of the line driver

2.1.1 Task Description

The objective of this task is to measure the rise and fall times of a signal generator and the open output of a line driver. The line driver (LM5134 and LMG1025) is supplied with +5V, and the function generator is set to output a rectangular signal with 5Vpp and a 2.5V DC offset.

2.1.2 Schematic

The measurement configuration consists of an Arbitrary Waveform Generator connected to the input of the line driver. The output of the driver is connected to an SMA connector via two $2.7\ \Omega$ series resistors, providing a total of $5.4\ \Omega$ series resistance. The voltage is detected using a $1\ \text{k}\Omega$ resistor connected to a second SMA connector, which alongside the $50\ \Omega$ input of the oscilloscope and a $50\ \Omega$ coaxial transmission line, forms a 1:21 voltage divider.

(Placeholder: Recreate or insert Figure 2-1 from the lab manual showing the block diagram of the measuring arrangement).

2.1.3 Formulas and Calculations

For this specific measurement, the rise and fall times are evaluated directly from the oscilloscope's transient voltage curves using the instrument's automatic measurement functions or cursors (typically measuring the transition from 10% to 90% of the signal amplitude). No additional complex mathematical derivations are required for this step.

2.1.4 Table(s) with measurement results

(Placeholder: Insert table here. Example format below)

Table 2.1: Rise and fall times of the signal generator and line driver.

Adjusted	Measured	Calculated
Test Point	Rise Time t_r (ns)	Fall Time t_f (ns)
Signal Generator	[Result]	[Result]
Line Driver (Open Output)	[Result]	[Result]

Notes: Function generator set to 5Vpp, 2.5V DC offset.

2.1.5 Curves & diagrams

(Placeholder: Insert high-resolution vector images of the oscilloscope plots showing the rise and fall times. Ensure cursor values and/or automatic measurement functions are visible in the plot as per guidelines).

2.1.6 Short discussion of measurement results

(Placeholder: Insert a short discussion comparing the rise and fall times of the signal generator versus the line driver. Include an error estimation and explain any deviations from the expected behavior without using general phrases).

2.2 Measurement B: Reflections of a coax cable with different loads

2.2.1 Task Description

An RG142 coaxial cable with a minimum length of 1 m is connected to the line driver output. The task is to terminate the end of the cable with different loads (Open, Short, 50 Ω , 100 Ω , 25 Ω , 10 nF capacitor, and 1 mH inductor) and measure the resulting line reflections. The transient voltage curve must be recorded to determine the time delay between reflection minima, the maximum voltage, and the steady-state asymptotic voltage. Using the measured propagation delay and physical cable length, the relative dielectric constant (ϵ_r) of the cable must be calculated.

2.2.2 Schematic

The basic schematic is identical to Measurement A, but the open end of the RG142 coaxial line is now terminated with various load impedances (Z_L).

(Placeholder: Insert the circuit diagram indicating the transmission line and the variable load attached at the far end).

2.2.3 Formulas and Calculations

The reflection factor ρ_0 at the load is determined by the load impedance Z_L and the characteristic impedance Z_w of the transmission line:

$$\rho_0 = \frac{Z_L - Z_w}{Z_L + Z_w} \quad (2.1)$$

The phase velocity v_p of the signal propagating through the transmission line is related to the speed of light in a vacuum ($c = 2.9979 \times 10^8$ m/s) and the relative dielectric constant ϵ_r :

$$v_p = \frac{c}{\sqrt{\epsilon_r}} \quad (2.2)$$

The time delay t measured between the incident wave and the reflected wave represents the round trip of the signal along the length of the cable l . The delay is:

$$t = \frac{2l}{v_p} \quad (2.3)$$

By substituting the phase velocity formula into the time delay formula, the dielectric constant ϵ_r can be calculated from the measured physical length and measured time delay:

$$\epsilon_r = \left(\frac{2l}{c \cdot t} \right)^2 \quad (2.4)$$

2.2.4 Table(s) with measurement results

(Placeholder: Insert table here. Example format below)

2.2.5 Curves & diagrams

(Placeholder: Insert oscilloscope plots for each of the 7 loads tested. Ensure trace averaging is used, and the time delay between reflection minima is clearly shown using cursors).

Table 2.2: Reflection characteristics and calculated parameters for various loads.

Load	t (ns)	V_{max} (V)	V_{steady} (V)	ρ_0	ϵ_r
Open	[Result]	[Result]	[Result]	[Result]	[Result]
Short	[Result]	[Result]	[Result]	[Result]	[Result]
50 Ω	[Result]	[Result]	[Result]	[Result]	[Result]
100 Ω	[Result]	[Result]	[Result]	[Result]	[Result]
25 Ω	[Result]	[Result]	[Result]	[Result]	[Result]
10 nF	[Result]	[Result]	[Result]	[Result]	[Result]
1 mH	[Result]	[Result]	[Result]	[Result]	[Result]

Notes: Physical length of RG142 cable $l = [\text{Measured Length}] \text{ m}$.

2.2.6 Short discussion of measurement results

(Placeholder: Analyze the transient responses. Discuss how the different terminations (resistive, capacitive, inductive) affected the reflection coefficient and the steady-state voltage. Provide error estimation for the calculated dielectric constant).

2.3 Measurement C: Determine cable length

2.3.1 Task Description

The goal is to determine the unknown physical length of a provided RG58 coaxial cable. This is achieved using Time Domain Reflectometry by measuring the propagation delay of a reflection and utilizing the relative dielectric constant (ϵ_r) determined in Measurement B.

2.3.2 Schematic

The measurement setup is the same as the previous reflectometry setups, but the standard RG142 cable is replaced with the RG58 coaxial cable of unknown length. The end of the cable is left open or shorted to ensure a strong reflection can be measured.

(Placeholder: Insert a brief schematic or block diagram showing the line driver connected to the "Unknown Length RG58 Cable").

2.3.3 Formulas and Calculations

Using the time delay t measured from the oscilloscope and the phase velocity v_p (derived from the ϵ_r calculated in Measurement B), the unknown length l of the cable can be calculated by rearranging the delay formula:

$$l = \frac{v_p \cdot t}{2} = \frac{c \cdot t}{2\sqrt{\epsilon_r}} \quad (2.5)$$

2.3.4 Table(s) with measurement results

(Placeholder: Insert table here. Example format below)

Table 2.3: Determination of the unknown RG58 cable length.

Assumed ϵ_r	Time Delay t (ns)	Calculated Length l (m)
Value from B	Result	Result

2.3.5 Curves & diagrams

(Placeholder: Insert the oscilloscope plot showing the incident and reflected wave used to measure the time delay for the unknown RG58 cable).

2.3.6 Short discussion of measurement results

(Placeholder: Discuss the calculated length. Estimate the uncertainty in the measurement based on the cursor resolution of the oscilloscope and the potential error carried over from the ϵ_r calculated in Measurement B).

3 S-Parameters

3.1 Coaxial Filters

3.1.1 Task Description

The task is to calibrate the network analyser (transmission and reflection for all 4 ports) and measure the 2-port parameters for coaxial filters. Based on these measurements, the order of the filters and their characteristics must be determined. Finally, the reflection coefficients are to be displayed as a Smith chart, and the transmission parameters must be displayed as logarithmic magnitude and phase.

(Note: Section 3.1.2 Schematic is excluded as the measurement involves connecting the coaxial filters directly to the network analyser, which cannot be represented by a unique standard symbol and does not require a complex circuit diagram. Section 3.1.3 Formulas and Calculations is also excluded because no derivations or manual calculation steps are required for this specific measurement.)

3.1.2 Table(s) with measurement results

(Placeholder: We will insert the measurement table here once you provide the data. As per the guidelines, the columns must be arranged from left to right containing adjusted, measured, and calculated values, separated clearly. Variables will be written in italics, constants and units upright, and units will not be put in square brackets.)

Table 3.1: Measured 2-port S-parameters for the coaxial filters

Adjusted	Measured	Calculated
Frequency / Hz	S_{11} , S_{21} , etc.	[Any calculated derivations if applicable]

3.1.3 Curves & diagrams

(Placeholder: We will insert your network analyser plots here. Remember that the screen images must not be copied with the snipping tool; they should be high-resolution vector images or direct PNG exports. The names of the students must be inserted on the instrument pictures, and trace labels with self-explanatory names should be used.)

Figure 3.1: Reflection coefficients of the coaxial filters displayed in a Smith chart. [Insert Smith chart plot here showing S_{11} and/or S_{22}]

Figure 3.2: Transmission parameters of the coaxial filters displayed as logarithmic magnitude and phase. [Insert Bode plot here showing S_{21} and/or S_{12}]

3.1.4 Short discussion of measurement results

(Placeholder text to be updated with your actual results): Considering the measured transmission parameters (logarithmic magnitude and phase), the filter exhibits a [insert characteristic: e.g., low-pass, high-pass, band-pass] characteristic. The order of the filter can be determined by observing the roll-off rate (attenuation slope) in the stopband of the logarithmic magnitude plot. [Insert specific roll-off rate, e.g., -40 dB/decade, and conclude

the order, e.g., 2nd order]. Furthermore, the Smith chart displaying the reflection coefficients indicates [insert observations about impedance matching, e.g., the filter is well-matched to $50\ \Omega$ in the passband as the curve approaches the center of the Smith chart, but highly reflective in the stopband].

3.2 Microstrip Lines

3.2.1 Task Description

The task is to measure the transmission coefficient (S_{21}) and the reflection coefficient (S_{11}) for different microstrip lines using a network analyser.

3.2.2 Schematic

(Placeholder: Insert your measurement setup or marked measuring points here. The network analyser does not need to be drawn in detail, but the measuring points must be clearly marked.)

(Note: Section 3.2.3 Formulas and Calculations is excluded as the task relies purely on direct network analyser measurements, and no specific analytical derivations are required.)

3.2.3 Table(s) with measurement results

(Placeholder: Insert your tabulated data for the microstrip lines here. Ensure the columns are ordered as adjusted, measured, and calculated from left to right, clearly separated.)

Table 3.2: Measured transmission and reflection coefficients for the microstrip lines

Adjusted	Measured	Calculated
Frequency / Hz	S_{11} / dB or linear	[Calculated Impedance or VSWR if applicable]
Frequency / Hz	S_{21} / dB or linear	[Any calculated derivations]

3.2.4 Curves & diagrams

(Placeholder: Insert your high-resolution vector graphics or PNGs of the reflection and transmission coefficients here. Ensure trace labels have self-explanatory names and the axes are properly labeled.)

Figure 3.3: Measured reflection (S_{11}) and transmission (S_{21}) coefficients of the microstrip lines over frequency. [Insert Plot Here]

3.2.5 Short discussion of measurement results

(Placeholder text to be updated with your actual results): Considering the plotted diagrams, one can observe the different reflection and transmission coefficients for all the measured microstrip lines. The reflection coefficient curves exhibit [insert number] minima, indicating that at these specific frequencies the impedance is approximately [insert impedance value, e.g., $50\ \Omega$]. Furthermore, the transmission coefficients change [describe behavior, e.g., linearly/non-linearly] with the frequency, showing a decrease of approximately [insert value] dB per [insert frequency range], which illustrates the frequency-dependent losses of the specific microstrip lines. [Insert any error estimation or explanations for deviations here].

3.3 3-Port Circulator

3.3.1 Task Description

The objective of this task is to measure the complete S-parameter matrix of a 3-port circulator using a network analyser and compare the measured characteristics against the ideal theoretical behaviour of a circulator.

3.3.2 Schematic

(Placeholder: Insert the block diagram or standard symbol of the 3-port circulator with its ports clearly labeled 1, 2, and 3. As per the guidelines, the network analyser itself does not need to be drawn, but the measuring points connecting to the circulator ports must be marked clearly.)

3.3.3 Formulas and Calculations

A circulator is an asymmetric, non-reciprocal component. According to the laboratory manual, the ideal scattering matrix (S-matrix) of a perfectly matched, lossless 3-port circulator is defined as:

$$S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad (3.1)$$

Network analysers typically measure and display S-parameters in decibels (dB), defined by the relationship:

$$S_{\text{dB}} = 20 \log(|S|) \quad (3.2)$$

To compare the measured results with the linear voltage ratios of the ideal S-matrix, the measured dB values must be converted back to linear magnitudes using the following calculation:

$$|S_{\text{linear}}| = 10^{\frac{S_{\text{dB}}}{20}} \quad (3.3)$$

3.3.4 Table(s) with measurement results

(Placeholder: Insert your tabulated data here. The columns must be clearly separated and ordered as adjusted, measured, and calculated. Variables must be in italics, units upright, and units must not be in square brackets.)

Table 3.3: Measured and calculated S-parameters of the 3-port circulator at optimal frequency

Adjusted	Measured	Calculated
Frequency / Hz	S_{11} / dB	$ S_{11} $ linear magnitude
Frequency / Hz	S_{21} / dB	$ S_{21} $ linear magnitude
Frequency / Hz	[... Continue for all 9 S-parameters]	[... Continue for linear calculations]

3.3.5 Curves & diagrams

(Placeholder: Insert the high-resolution vector images or PNG exports from the network analyser here. Ensure the images show the S-parameters over the measured frequency span to illustrate the frequency dependency. The names of the students must be inserted on the instrument pictures, and trace labels with self-explanatory names must be used.)

Figure 3.4: Measured S-parameters of the 3-port circulator over frequency. [Insert Plot Here]

3.3.6 Short discussion of measurement results

(Placeholder text to be updated with your actual results): Considering the ideal S-matrix, the main diagonal entries (S_{11} , S_{22} , S_{33}) represent the reflection coefficients at each port, which should ideally be 0. The off-diagonal entries represent the transmission coefficients (e.g., S_{21} represents the wave transmitted from port 1 to port 2). In the ideal matrix, elements S_{13} , S_{21} , and S_{32} are 1, dictating that power flows in a specific clockwise or counter-clockwise circle between the ports.

Based on the measured curves, the circulator is highly frequency-dependent and exhibits its optimal characteristic at a frequency of approximately [insert frequency] Hz. At this frequency, the linear S-matrix was calculated to be:

[Insert your calculated 3x3 linear matrix here]

While the measured matrix does not perfectly match the ideal theoretical matrix—exhibiting an insertion loss of approximately [insert value] dB in the forward direction and a finite isolation of [insert value] dB in the reverse direction—the characteristic circular power flow of the component is clearly visible. [Insert an error estimation or explanation regarding any specific deviations from the ideal behaviour, component tolerances, or limits of the measurement setup here as required by the guidelines].

3.4 3-Port Resistive Power Divider

3.4.1 Task Description

The task is to measure the S-parameters of a 3-port resistive power divider using a network analyser. The objective is to verify that the power divider matches all three ports to the characteristic impedance and exhibits a total loss of 6 dB. Furthermore, the measurement should determine whether there is any isolation between output ports 2 and 3.

3.4.2 Schematic

(Placeholder: Insert the circuit diagram or block symbol of the 3-port resistive power divider here. Ensure the three ports are clearly marked as Port 1, Port 2, and Port 3. As per the guidelines, measuring points must be clearly marked.)

3.4.3 Formulas and Calculations

A completely matched, reciprocal 3-port network cannot be lossless. A resistive power divider uses resistors to match all three ports simultaneously to the characteristic impedance (Z_0). Because of the reciprocal and symmetrical nature of this specific resistive network, the theoretical linear S-parameter matrix is expected to be:

$$S = \frac{1}{2} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad (3.4)$$

Network analysers display these values in decibels (dB). The theoretical transmission loss of $\frac{1}{2}$ corresponds to:

$$S_{\text{dB}} = 20 \log(0.5) \approx -6 \text{ dB} \quad (3.5)$$

This explains the theoretical 6 dB total loss between the input and any output port.

3.4.4 Table(s) with measurement results

(Placeholder: Insert your tabulated data here. The columns must be arranged to contain adjusted, measured, and calculated values from left to right, with a clear visible separation.)

Table 3.4: Measured S-parameters of the 3-port resistive power divider

Adjusted	Measured	Calculated
Frequency / Hz	S_{11}, S_{22}, S_{33} / dB	Linear reflection coefficients
Frequency / Hz	$S_{21}, S_{31}, S_{12}, S_{13}$ / dB	Linear transmission coefficients
Frequency / Hz	S_{23}, S_{32} / dB	Linear isolation coefficients

3.4.5 Curves & diagrams

(Placeholder: Insert high-resolution vector images or PNG exports from the network analyser showing the S-parameters. Ensure main and auxiliary grids are switched on, and trace labels are self-explanatory.)

Figure 3.5: Measured S-parameters (transmission, reflection, and isolation) of the 3-port resistive power divider. [Insert Plot Here]

3.4.6 Short discussion of measurement results

(Placeholder text to be updated with your actual results): Based on the measurement results, the reflection coefficients (S_{11} , S_{22} , S_{33}) are approximately [insert value] dB, which confirms that all three ports are [well-matched / poorly matched] to the characteristic impedance. The forward transmission coefficients (S_{21} and S_{31}) were measured at approximately [insert value] dB, which [confirms / deviates from] the theoretical total loss of 6 dB.

Regarding the relationship between ports 2 and 3, the measured parameters S_{23} and S_{32} are approximately [insert value] dB. This demonstrates that there is [no isolation / high isolation] between port 2 and port 3, as the signal is [freely transmitted / blocked] between these two output ports. [Insert an error estimation regarding the component tolerances of the internal resistors or measurement accuracy to explain any slight deviations from the ideal 6 dB or 0 dB values].

3.5 Antenna Reflection Coefficients

3.5.1 Task Description

The task is to measure the reflection coefficient (S_{11}) of two different antennas (specifically a 2.4 GHz antenna and a 915 MHz antenna) using a network analyser. The measurement results must be displayed both as a Smith chart and as a logarithmic magnitude plot over frequency.

3.5.2 Schematic

(Note: As the measurement simply involves connecting the antennas directly to the network analyser's port, a full circuit diagram is not strictly necessary as the network analyser cannot be represented by a unique standard symbol. If you have a photograph of the physical antennas or a block diagram marking the measuring point at the antenna feed, it can be inserted here.)

3.5.3 Formulas and Calculations

The reflection factor ρ (which corresponds to the S_{11} parameter measured by the network analyser) at the point of the load is defined as the ratio of the backward to the forward voltage wave. It relates the antenna's input impedance (Z_L) to the characteristic impedance of the measurement system (Z_w , typically 50 Ω):

$$\rho = \frac{Z_L - Z_w}{Z_L + Z_w} \quad (3.6)$$

The network analyser typically displays the logarithmic magnitude of the reflection coefficient in decibels, which is calculated as:

$$S_{11}[\text{dB}] = 20 \log(|S_{11}|) \quad (3.7)$$

3.5.4 Table(s) with measurement results

(Placeholder: Insert your tabulated data for both antennas here. Remember to clearly separate the columns for adjusted, measured, and calculated values.)

Table 3.5: Measured reflection coefficients at the target resonance frequencies of the antennas

Adjusted	Measured	Calculated
Frequency / Hz	2.4 GHz Antenna S_{11} / dB	Calculated VSWR or Impedance Z_L / Ω
Frequency / Hz	915 MHz Antenna S_{11} / dB	Calculated VSWR or Impedance Z_L / Ω

3.5.5 Curves & diagrams

(Placeholder: Insert high-resolution vector images or PNG exports from the network analyser here. Ensure you have one plot showing the Smith chart and one showing the logarithmic magnitude (Bode plot) for the measured frequency spans. Trace labels with self-explanatory names must be used.)

Figure 3.6: Measured reflection coefficients of the 2.4 GHz and 915 MHz antennas displayed as a Smith chart. [Insert Smith Chart Plot Here]

Figure 3.7: Measured reflection coefficients (logarithmic magnitude) of the 2.4 GHz and 915 MHz antennas. [Insert Log Magnitude Plot Here]

3.5.6 Short discussion of measurement results

(Placeholder text to be updated with your actual results): Based on the logarithmic magnitude plot, the 2.4 GHz antenna exhibits its minimum reflection coefficient of approximately [insert value] dB at exactly [insert

measured optimal frequency] GHz. Similarly, the 915 MHz antenna shows its optimal matching with an S_{11} of [insert value] dB at [insert measured optimal frequency] MHz. This indicates that at these specific frequencies, the antennas are [well-matched / poorly matched] to the $50\ \Omega$ system impedance, minimizing the reflected wave. By analyzing the Smith chart, the normalized input impedance of the antennas at their respective resonance frequencies can be read directly. At resonance, the imaginary part (reactance) should ideally be zero. The Smith chart shows that at the target frequencies, the curves pass [close to / far from] the matching point in the center ($r = 1, x = 0$). [Insert any error estimation or commentary regarding the usable bandwidth (e.g., the frequency range where $S_{11} < -10$ dB) or deviations caused by the surrounding environment/measurement setup here].

3.6 Unmatched Antenna and Matching Network

3.6.1 Task Description

The first objective of this task is to measure the reflection coefficient (magnitude and phase) of an unmatched antenna at a given frequency. Based on this measurement, an impedance matching network consisting of a transmission line and a series capacitor must be calculated to match the antenna to the $50\ \Omega$ system. The solution must be derived using two methods: an analytical calculation and a graphical solution using the Smith chart. Finally, the calculated matching network is to be implemented (if applicable) or verified to observe the final matched reflection coefficient.

3.6.2 Schematic

(Placeholder: Insert two block diagrams here. The first should show the unmatched antenna connected to the network analyser port. The second should show the equivalent circuit of the matching network: the antenna (Z_A) connected to a transmission line of length l_e , followed by a series capacitor C connecting to the $50\ \Omega$ measurement port.)

3.6.3 Formulas and Calculations

The unmatched antenna presents a load impedance Z_A which can be determined from the measured reflection coefficient $\rho_A = |\rho_A|e^{j\delta}$. The normalized load impedance is $z_A = Z_A/Z_w$.

a) Analytical Solution: To match the antenna using a lossless transmission line (length l_e) and a series reactive element (jX_s), the reflection coefficient is transformed along the line towards the generator:

$$\rho_+(-l_e) = \rho_A e^{-j\frac{4\pi}{\lambda}l_e} \quad (3.8)$$

The normalized impedance at this point is $z_+(-l_e) = \frac{1-\rho_+(-l_e)}{1+\rho_+(-l_e)}$. Adding the series reactance jx_s , the total normalized impedance must equal the matching condition ($1 + j0$):

$$jx_s + z_+(-l_e) - 1 \stackrel{!}{=} 0 \quad (3.9)$$

By separating this equation into real and imaginary parts, the two unknown quantities (x_s and l_e) can be calculated. The normalized reactance x_s is found via:

$$x_s = \pm \sqrt{\frac{1 - |\rho_A|^2}{|\rho_A|^2}} \quad (3.10)$$

Since the matching network specifically requires a capacitor, the negative solution for x_s must be chosen. The

actual capacitance C is then calculated from the denormalized reactance $X_s = Z_w x_s$:

$$X_s = -\frac{1}{\omega C} \implies C = -\frac{1}{\omega X_s} \quad (3.11)$$

b) Smith Chart Solution: In the Smith chart, the initial normalized impedance z_A is plotted. Moving along the transmission line corresponds to rotating clockwise (towards the generator) along the constant $|\rho_A|$ circle. The rotation stops when this circle intersects the matching circle (the $r = 1$ circle). The distance rotated along the outer wavelength scale determines the required line length l_e . At this intersection, the impedance has the form $1 + jx$. To achieve perfect matching, a series capacitor providing a normalized reactance of $-jx$ must be added to move the point to the centre of the Smith chart ($1 + j0$).

3.6.4 Table(s) with measurement results

(Placeholder: Insert your tabulated data here. The table should clearly contrast the unmatched state with the calculated matching network values.)

Table 3.6: Measurement of the unmatched antenna and calculated matching network parameters

Adjusted	Measured	Calculated
Frequency / Hz	Unmatched ρ_A Mag / dB	Unmatched Z_A / Ω
Frequency / Hz	Unmatched ρ_A Phase / °	Required line length l_e / m
Frequency / Hz	Matched S_{11} / dB (if measured)	Required Capacitance C / F

3.6.5 Curves & diagrams

(Placeholder: Insert your Smith chart construction here showing the graphical matching process. Additionally, insert a logarithmic magnitude plot from the network analyser comparing the S_{11} of the antenna before and after matching.)

Figure 3.8: Graphical solution of the matching network (line + capacitor) constructed in the Smith chart.

Figure 3.9: Measured reflection coefficient (logarithmic magnitude) of the antenna before and after applying the matching network.

3.6.6 Short discussion of measurement results

(Placeholder text to be updated with your actual results): The initial measurement of the unmatched antenna at [insert frequency] yielded a reflection coefficient of [insert magnitude] dB and a phase of [insert phase]°, corresponding to an input impedance of $Z_A =$ [insert impedance] Ω . This indicates a [high/low] degree of mismatch.

Using both the analytical formulas and the graphical Smith chart method, the required matching network was determined. The graphical and analytical results [perfectly aligned / showed slight deviations due to reading inaccuracies], yielding a required transmission line length of $l_e =$ [insert length] m and a series capacitance of $C =$ [insert capacitance] pF. [If you physically implement or simulate the match: Upon applying the matching

network, the reflection coefficient dropped significantly to [insert value] dB at the target frequency, successfully validating the calculated impedance matching network].

3.7 Slide-Screw Tuner

3.7.1 Task Description

The final task is to use a mechanical slide-screw tuner to match the previously unmatched antenna and to measure the final reflection coefficient using the network analyser.

3.7.2 Schematic

(Placeholder: Insert the block diagram showing the network analyser connected to the slide-screw tuner, which is then connected to the antenna. You should also insert the mechanical construction and electrical equivalent circuit of the slide-screw tuner (as shown in Figure 3-1 of the lab manual) to illustrate how the device operates.)

3.7.3 Formulas and Calculations

A mechanical slide-screw tuner consists of a $50\ \Omega$ slabline and a movable capacitive/reflective probe, known as a slug. The impedance matching is achieved mechanically rather than through analytical calculations:

- When the slug is fully retracted, the tuner presents an ideal $50\ \Omega$ impedance.
- As the slug is lowered into the slabline (Y-direction), it interrupts the electric field and introduces a capacitance, which increases the magnitude of the reflection.
- As the slug travels longitudinally along the slabline (X-direction), it rotates the phase of the reflection.

By adjusting both the X and Y positions, it is possible to re-create arbitrary impedances to perfectly match the antenna without requiring discrete lumped components or fixed transmission lines.

3.7.4 Table(s) with measurement results

(Placeholder: Insert your tabulated data here comparing the unmatched and matched state. Ensure columns are ordered as adjusted, measured, and calculated.)

Table 3.7: Reflection coefficient of the antenna before and after applying the slide-screw tuner

Adjusted	Measured	Calculated
Frequency / Hz	Unmatched S_{11} / dB	Calculated VSWR before
Frequency / Hz	Matched S_{11} / dB	Calculated VSWR after

3.7.5 Curves & diagrams

(Placeholder: Insert your network analyser plots here. Include a Smith chart demonstrating the movement of the reflection coefficient during the tuning process, and a logarithmic magnitude plot showing the final matched S_{11} minimum at the target frequency. Trace labels must be self-explanatory.)

Figure 3.10: Smith chart illustrating the impedance matching process using the slide-screw tuner.

Figure 3.11: Measured final reflection coefficient (logarithmic magnitude) of the antenna matched with the slide-screw tuner.

3.7.6 Short discussion of measurement results

(Placeholder text to be updated with your actual results): The unmatched antenna initially exhibited a high reflection coefficient, indicating poor matching to the $50\ \Omega$ system. By observing the Smith chart on the network analyser in real-time, the slide-screw tuner was adjusted. Lowering the slug (Y-direction) altered the magnitude of the reflection, while sliding it along the line (X-direction) rotated the phase, steering the impedance curve directly toward the matching point in the center of the Smith chart ($r = 1, x = 0$).

The final measurement yielded a reflection coefficient of [insert matched value] dB at the target frequency of [insert frequency]. This demonstrates that the mechanical slide-screw tuner successfully matched the antenna to the system, practically validating the theoretical matching principles without the need for fixed capacitors or inductors. [Insert any error estimation regarding the mechanical sensitivity of the tuner or measurement stability here].

Figure 4.2 shows the schematic of the two additional inverting amplifier (LM4562) stages, connected to the output of the SC-filter circuit (second stage gain = -100, dual supply = $\pm 5V$, and SC-filter tuning frequency = 1MHz).

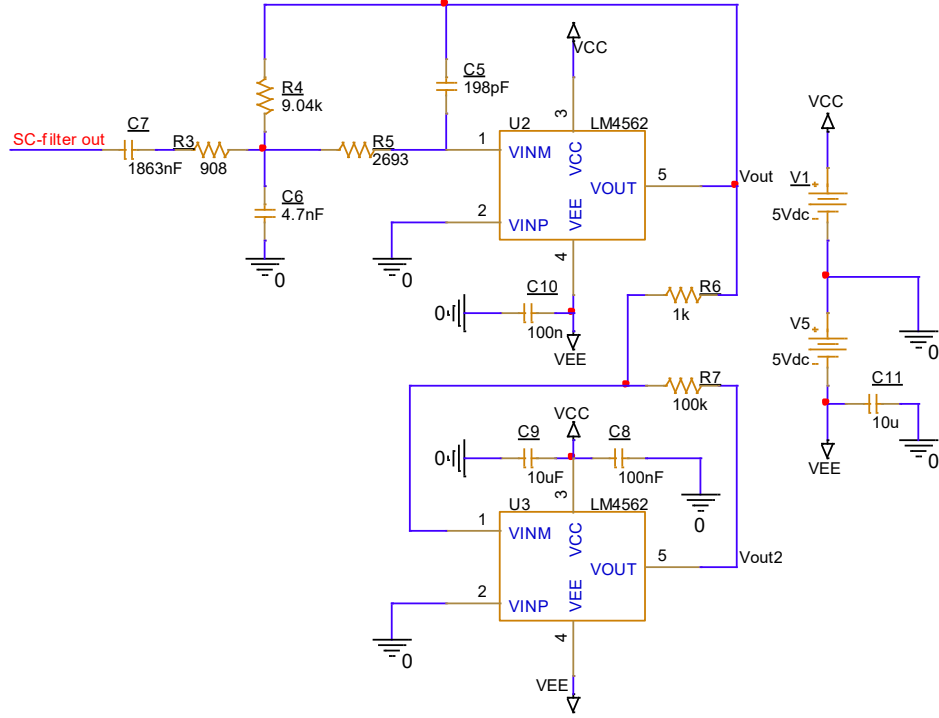


Figure 4.2: Schematic of the two-stage inverting amplifiers for RMS noise measurement.

4.3 Formulas and Calculations

The following formulae (4.1) and (4.2) were used to calculate the SNR and ENOB. [2].

$$SNR = 20 \log_{10} \left(\frac{U_{in} Gain}{\sqrt{2} U_n} \right) = 49.40 dB \quad (4.1)$$

$$ENOB = \frac{SNR - 1.76}{6.02} = 7.914 = 8 Bit \quad (4.2)$$

4.4 Tables with measurement results

Table 4.1 shows the calculated, measured and selected component values for the switched capacitor filter.

Components Resistors/Capacitors	Selected $k\Omega/nF$	Measured $k\Omega/nF$	Calculated $k\Omega/nF$
R_g	0.05	0.051	-
R_1	1	0.997	-
R_2	100	100.7	-
R_3	0.91	0.908	0.851
R_4	9.1	9.04	8.519
R_5	2.7	2.69	2.635
R_6	1	0.995	-
R_7	100	100.7	-
C_1	100	99	-
C_2	100	100	-
C_3	100	103	-
C_4	100	103	-
C_5	0.15	0.198	-
C_6	4.7	4.69	-
C_7	1500	1501	1863
C_8	100	100.09	-
C_9	10000	10030	-
C_{10}	100	101	-
C_{11}	10000	9950	-

Table 4.1: Measured values for the resistors and capacitors use for the switched capacitor filter, the pre-amplification stage, and the two inverting amplifier stages.

Adjusted	Measured
R_g / Ω	U_n / mV
51	125.16
100k	239.3

Table 4.2: Measured values for total noise voltage of the switched capacitor filter for 2 values of R_g .

4.5 Measurement curves and diagrams

Figure 4.3 shows the bode plot of the switched capacitor filter, along with the pre-amplification stage. Gain of the circuit at frequency values of 500kHz and 1MHz are illustrated here.

Tables 5.3 and 5.4 show the measured Gain values for both tuning frequencies.

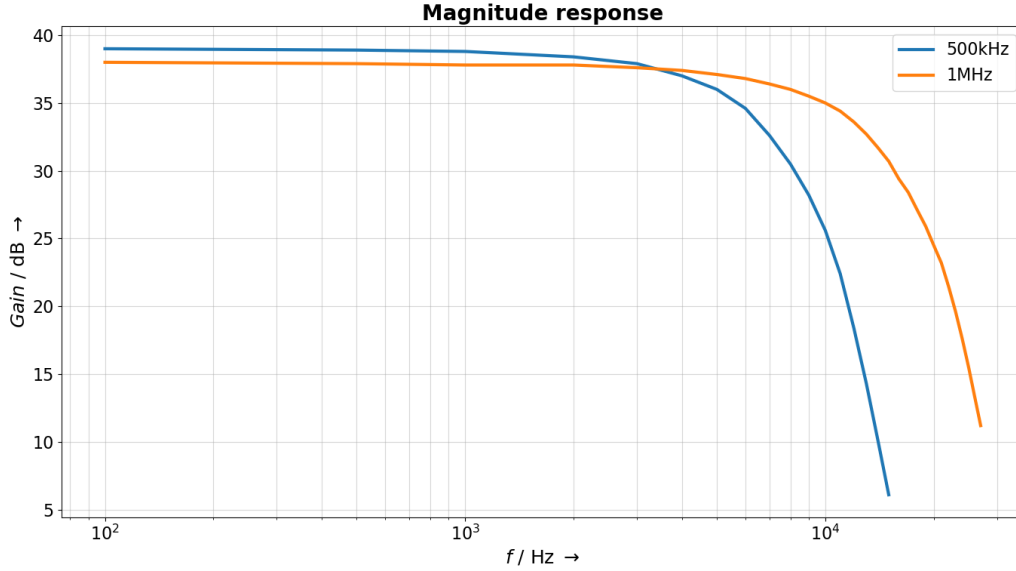


Figure 4.3: Magnitude response of the switched capacitor filter ($U_{in}=20\text{mVpp}$).

Figure 4.4 shows the output waveform of the switched capacitor filter along with the pre-amplification stage. The input frequency is set near tuning frequency.

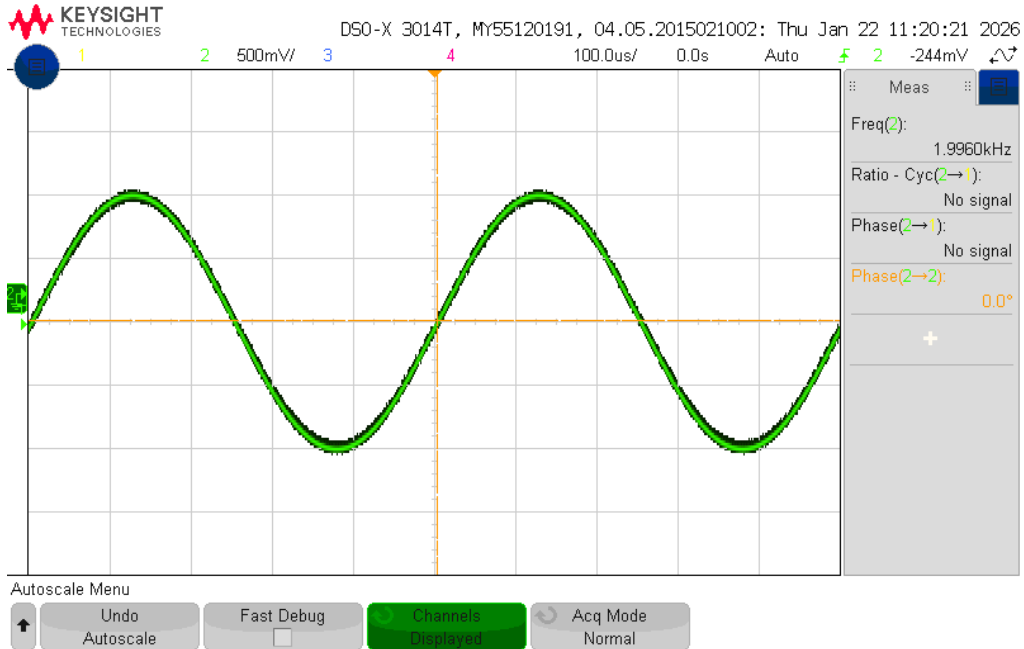


Figure 4.4: Output plot of the switched capacitor filter ($U_{in}=20\text{mVpp}$, $f=998\text{kHz}$)

Figure 4.5 shows the output voltage corresponding to a sine wave as input voltage. Once zoomed in, the switching frequency is visible in the output voltage characterized by the step-like appearance.

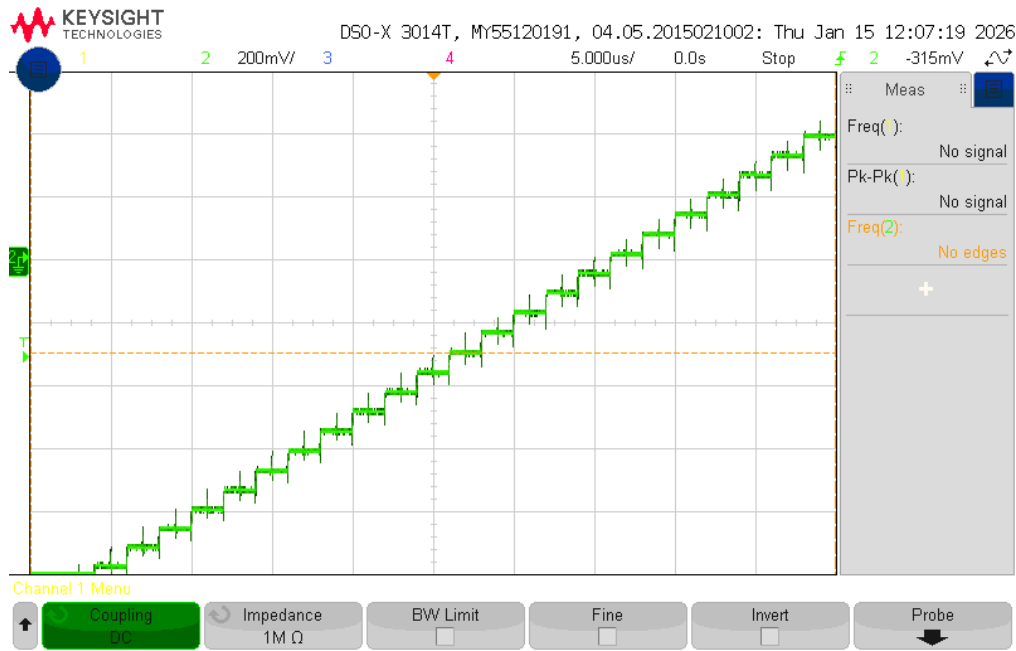


Figure 4.5: Resultant waveform depicting the switching frequency ($U_{in}=20\text{mVpp}$, $f=5\text{kHz}$)

Figure 4.6 shows the measurement of the total noise voltage at the output of the filter including the inverting amplifier stages. The readings are illustrated using the RMS function as recommended.



Figure 4.6: Resultant graph showing the output noise with $R_g=100\text{k}\Omega$.

4.6 Discussion

- Figure 4.3 shows the Bode plot of the SC filter. Two different tuning frequencies were used for the measurements 500 kHz and 1 MHz. As expected for a switched capacitor filter, the lower the clock frequency, the attenuation begins at a lower cut-off frequency.
- Figure 4.4 shows the output voltage when the tuning frequency is close to the input frequency.
- Figure 4.5 shows a sine wave with $\hat{u}=20\text{mVpp}$ and $f=998\text{kHz}$ as the input signal. It can be seen that the output is an amplified sine wave with a frequency of 1.996 kHz. This frequency is the difference between the set input frequency and the clock frequency (1MHz). When the input is close to the clock frequency, sampling creates a low “difference” (alias/beat) frequency in the output.
- Figure 4.6 depicts the noise measurement using the RMS function.
- The SNR and ENOB values were also calculated (see 4.1 and 4.2).

5 Appendix

5.1 Calculations for the active band pass filter

Calculation of the transfer function of one filter stage:

$$U_{R2} = -U_{out} \quad (5.1)$$

$$U_{R3} = U_{C2} \quad (5.2)$$

$$I_{R2} = \frac{U_{R2}}{R_2} \quad (5.3)$$

$$U_{C2} = I_{R2} \cdot Z_2 = \frac{U_{R2}}{sR_2C_2} \quad (5.4)$$

$$U_{C1} = U_{C2} + U_{R2} = U_{R2} + \frac{U_{R2}}{sR_2C_2} \quad (5.5)$$

$$I_{C1} = \frac{U_{C1}}{Z_{C1}} = \left(U_{R2} + \frac{U_{R2}}{sR_2C_2} \right) \cdot sC_1 \quad (5.6)$$

$$I_{R3} = \frac{U_{R3}}{R_3} = \frac{U_{R2}}{sR_2C_2R_3} \quad (5.7)$$

$$I_{R1} = I_{C1} + I_{C2} + I_{R3} = sC_1U_{R2} + \frac{sC_1U_{R2}}{R_2C_2} + \frac{U_{R2}}{R_2} + \frac{U_{R2}}{sR_2C_2R_3} \quad (5.8)$$

$$C_1 = C_2 = C \quad (5.9)$$

$$U_{in} = U_{R3} + U_{R1} = R_1 \cdot I_{R1} + \frac{U_{R2}}{sR_2C} = \quad (5.10)$$

$$U_{R2} \left(\frac{1}{sR_2C} + \frac{2R_1}{R_2} + sCR_1 + \frac{R_1}{sR_2R_3C} \right) = \quad (5.11)$$

$$-U_{out} \frac{R_3 + 2sR_1R_3C + s^2R_1R_2R_3C^2 + R_1}{sR_2R_3C} \quad (5.12)$$

$$H(s) = \frac{U_{out}}{U_{in}} = - \frac{\frac{sR_2R_3C}{R_1+R_3}}{1 + \frac{2R_1R_3}{R_1+R_3}sC + \frac{R_1R_2R_3}{R_1+R_3}s^2C^2} \quad (5.13)$$

$$H(P) = - \frac{\frac{R_2R_3C}{R_1+R_3}\omega_0P}{1 + \frac{2R_1R_3}{R_1+R_3}\omega_0PC + \frac{R_1R_2R_3}{R_1+R_3}\omega_0^2P^2C^2}$$

Calculation the get the factor α :

The denominator of the function ?? is the following.

$$D(s) = [1 + \frac{\alpha s}{Q_i} + (\alpha s)^2] \cdot [1 + \frac{1}{Q_i} \left(\frac{s}{\alpha} \right) + \left(\frac{s}{\alpha} \right)^2] \quad (5.14)$$

$$= 1 + s \left(\frac{\alpha}{Q_i} + \frac{1}{Q_i\alpha} \right) + s^2 \left(\alpha^2 + \frac{1}{Q_i^2} + \frac{1}{\alpha^2} \right) + s^3 \left(\frac{\alpha}{Q_i} + \frac{1}{Q_i\alpha} \right) + s^4 \quad (5.15)$$

Coefficient comparison:

$$s^0 : 1 = 1 \quad (5.16)$$

$$s^1 : \frac{\alpha}{Q_i} + \frac{1}{Q_i \alpha} = \frac{\alpha}{b} \Delta \Omega \quad (5.17)$$

$$s^2 : \alpha^2 + \frac{1}{Q_i^2} + \frac{1}{\alpha^2} = 2 + \frac{\Delta \Omega^2}{b} \quad (5.18)$$

$$s^3 : \frac{\alpha}{Q_i} + \frac{1}{Q_i \alpha} = \frac{\alpha}{b} \Delta \Omega \quad (5.19)$$

$$s^4 : 1 = 1 \quad (5.20)$$

The equation ?? can be derived from s^1 or s^3 and must be substituted in the coefficient equation of s^2 as follows.

$$\alpha^2 + \left[\frac{\alpha \Delta \Omega a}{b(1 + \alpha^2)} \right] + \frac{1}{\alpha^2} - 2 - \frac{(\Delta \Omega)^2}{b} = 0 \quad (5.21)$$

The equation 5.21 was solved using Wolfram Alpha 5.1

Further necessary values for the calculations are:

$$Q = \frac{f_m}{f_u - f_l} = 4.47 \quad (5.22)$$

$$\Delta \Omega = \frac{1}{Q} \quad (5.23)$$

$$a = \sqrt{2}; b = 1 \quad (5.24)$$

$$a \text{ and } b \text{ are given for a Butterworth filter} \quad (5.25)$$

The chosen real solutions is $\alpha = 1.0826$

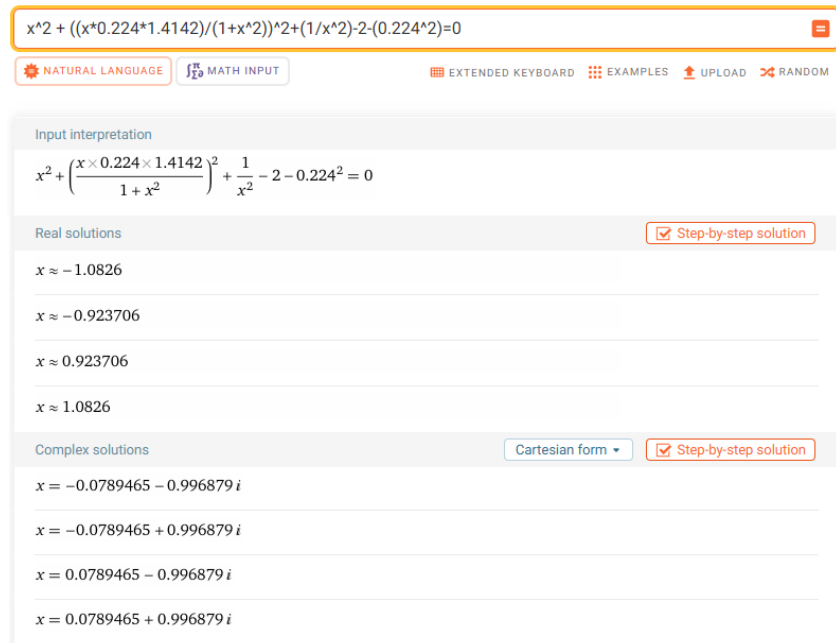


Figure 5.1: computing factor α

5.2 Measurement data for the active band pass filter

The following table shows the frequency response of the active band pass filter.

frequency Hz	magnitude dB	phase degree
3006	3.09	150
3500	13.5	118.36
3600	16	106
3700	17.9	90.52
3799	20	70.49
3901	21.7	48.61
4000	22.3	24.36
4099	22.5	3.83
4199	22.4	-18.4
4300	22.1	-38.66
4350	21.9	-49.27
4399	21.4	-58.44
4498	19.9	-80.67
4600	18.2	-98.24
4697	16.4	-108.74
4799	14.1	-118.33
4898	12.2	-126.8
5000	10.7	-132.15
5099	9.1	-136.27
5191	7.8	-140.66
5300	6.6	-142.73
5396	5.2	-146.02
5510	4.1	-148.46

Table 5.1: Measured frequency response of the 2nd order active band pass filter

5.3 Calculations for the ADC-Driver and anti-aliasing filter

These formulas can be found in the chapter *Active 2nd Order Low Pass (Multifeedback)* of [3]. From these equations the equations for the component values can be constructed.

$$H_0 = -\frac{R_2}{R_1} \quad (5.26)$$

$$a = \omega_C C_1 \left[R_2 + R_3 + \frac{R_2 R_3}{R_1} \right] \quad (5.27)$$

$$b = \omega_C^2 C_1 C_2 R_2 R_3 \quad (5.28)$$

5.4 Measurement data for the ADC-Driver and anti-aliasing filter

The following table shows the frequency response of the ADC-Driver and anti-aliasing filter.

frequency (Hz)	magnitude (dB)	phase (degree)
10	4.5	-108.4
30	13.6	-109.3
50	17.6	-115.9
80	20.6	-125.8
100	21.8	-131.2
200	23.9	-147.9
500	24.9	-163.7
800	25.1	-170.2
1000	25.1	-171.4
2000	25.2	-171.6
5000	25.2	-186.5
8000	25.2	-192.9
10000	25.2	-198.3
15000	25	-210
20000	24.7	-222
30000	23.7	-250.3
40000	22	-272.9
50000	19.8	-290.5
80000	13	-320.8
100000	9.4	-331.2
150000	2.7	-351.3

Table 5.2: Measured frequency response of the single-ended to differential driver

5.5 Measurement data for the switched capacitor filter

Table 5.3 show the magnitude response of the filter at 1MHz. The phase values were observed shifting around 0 in minimal quantities and hence this measurement has been excluded.

frequency (Hz)	magnitude (dB)
100	38.0
500	37.9
1000	37.8
2000	37.8
3000	37.6
4000	37.4
5000	37.1
6000	36.8
7000	36.4
8000	36.0
9000	35.5
10000	35.0
11000	34.4
12000	33.6
13000	32.7
14000	31.7
15000	30.7
16000	29.4
17000	28.4
18000	27.1
19000	25.9
20000	24.5
21000	23.2
22000	21.4
23000	19.6
24000	17.6
25000	15.5
26000	13.3
27000	11.2

Table 5.3: Measured frequency response of the switched capacitor filter at 1MHz.

frequency (Hz)	magnitude (dB)	phase (degree)
100	39.0	0.0
500	38.9	0.1
1000	38.8	0.3
2000	38.4.6	0.1
3000	37.9	0.0
4000	37.0	0.74
5000	36.0	1.67
6000	34.6	1.93
7000	32.6	2.3
8000	30.5	2.5
9000	28.2	3.0
10000	25.6	3.2
11000	22.4	2.0
12000	18.4	1.9
13000	14.3	2
14000	10.1	2.2
15000	6.1	2.5

Table 5.4: Measured frequency response of the switched capacitor filter at 500kHz.

6 Instrument device list

Device	Manufacturer	Model	Serialnumber
Oscilloscope	Keysight	InfiniiVision DSOX3014T	9964288
Digital Multimeter	Keysight	34450A	9964290
Waveform Generator	Keysight	33500B	9951296
Power Supply	Thurlby	PL330QMD	9949266

Table 6.1: Instrument device list

Graz, July 19, 2026

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Graz, July 19, 2026

Juanith Mathew Thomas

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